

Basic Principles of Medicine 2  
Module: Digestive System & Metabolism | Lecture 9

# Imaging of the Gastrointestinal System

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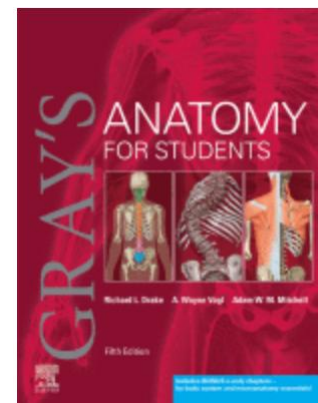


# Objectives

- |                                |                                                                                                                                        |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| SOM.MK.1.BMP2.3.DM.1.ANAT.1141 | Identify the various parts of the GI tract and their blood supply in CT, MRI and radiological images.                                  |
| SOM.MK.1.BMP2.3.DM.1.ANAT.1178 | Identify the liver, pancreas and spleen, their blood supply and other adjacent abdominal structures in CT, MRI and radiological images |
| SOM.MK.1.BPM2.3.DM.1.ANAT.1179 | Describe the anatomical location, mechanism and clinical sequences of the insertion of a nasogastric tube.                             |

# Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 5th ed. Elsevier Churchill Livingstone.
- Online Textbook: **Chapter 4**: [Click here](#)
- Pay special attention to the **green (print text)/blue (ebook)** “In the Clinic” boxes for the clinical correlations for this topic



# Introduction

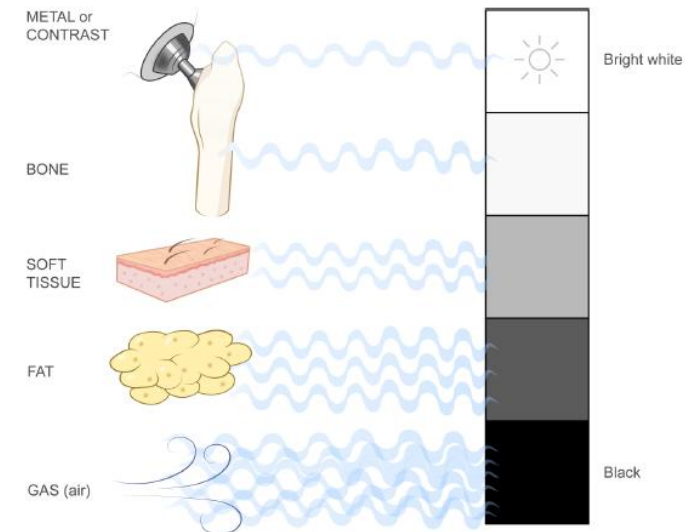
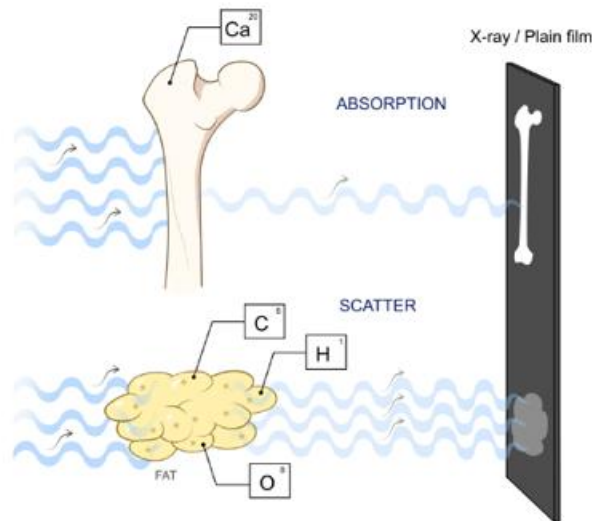
Imaging of the abdomen is now largely performed utilizing

- Computed Tomography (CT)
- Ultrasound (US)
- Magnetic Resonance Imaging (MRI)

However, many patients still have “plain films” (X-Ray) of the abdomen as a **first step** before other imaging studies are performed or as a **method of following up** on findings demonstrated by other modalities.

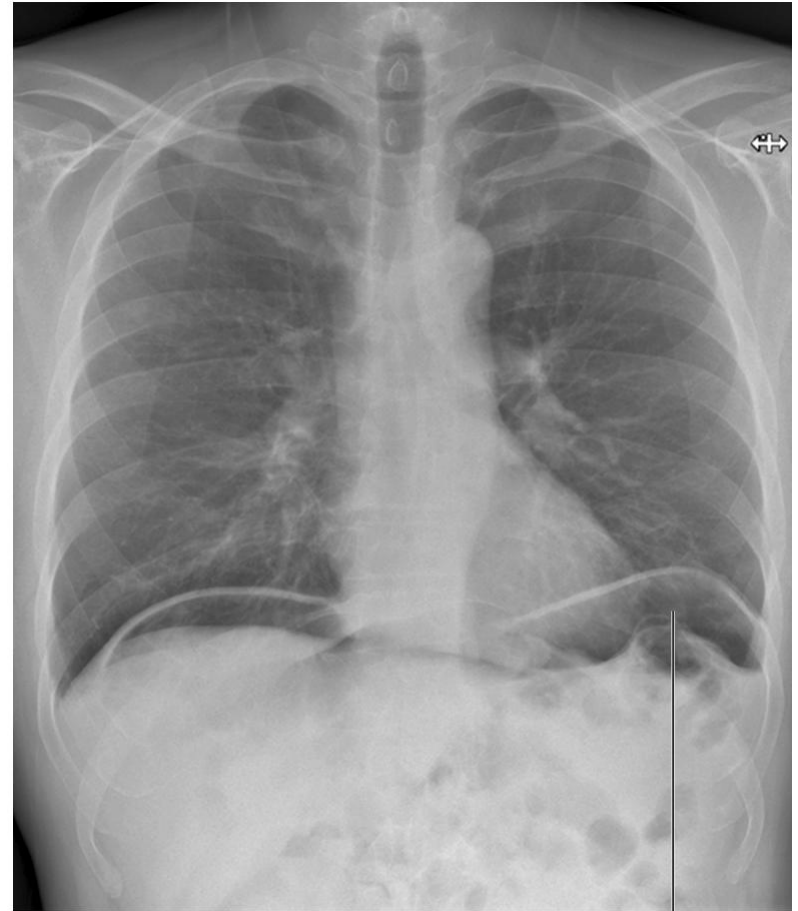
# Abdominal X-ray (AXR)

- Still commonly undertaken in the acute abdomen where perforation or acute bowel obstruction is suspected.
- Particularly used in the elderly where acute abdominal signs and symptoms are often nonspecific.
- Example indications for AXR include:
  - Abdominal pain where there is a suspicion of perforation or obstruction.
  - Foreign body ingestion
  - Postoperatively where there is a suspicion of ileus (*Review MSK*)



# Case I

A 61-year-old woman comes to the physician because of persistent abdominal discomfort and significant weight loss. She has a 2-month history of burning pain in the epigastric region of the abdomen and chest that radiates to the back. Her pain worsens after taking aspirin and drinking coffee and is relieved after taking antacids. She has lost 10 pounds in 2 months due to decreased dietary intake because of bloating, feelings of early satiety (fullness), and stomachaches between meals. Patient also reports doubling her NSAID intake due to increased knee pain from arthritis. Patient looks pale on observation.

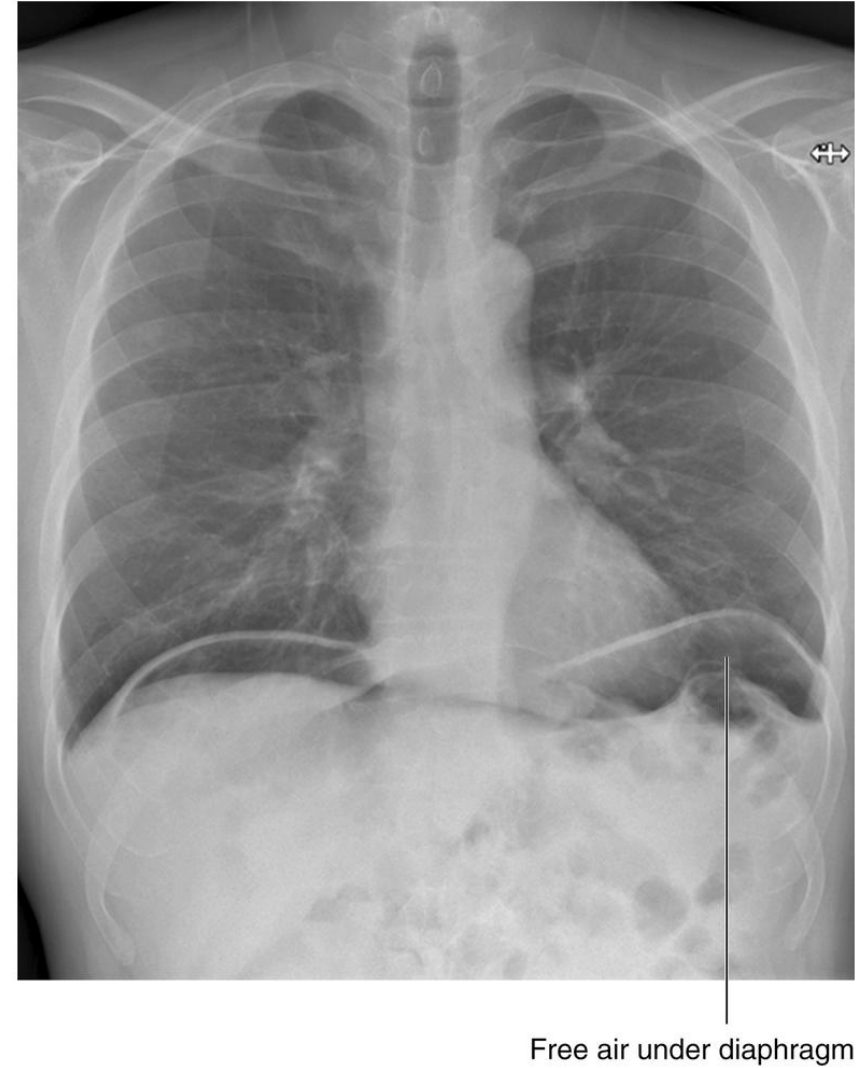


Free air under diaphragm

Chest AP upright radiograph. Free intraperitoneal air. The right and left hemidiaphragms are elevated owing to bilateral subdiaphragmatic air.

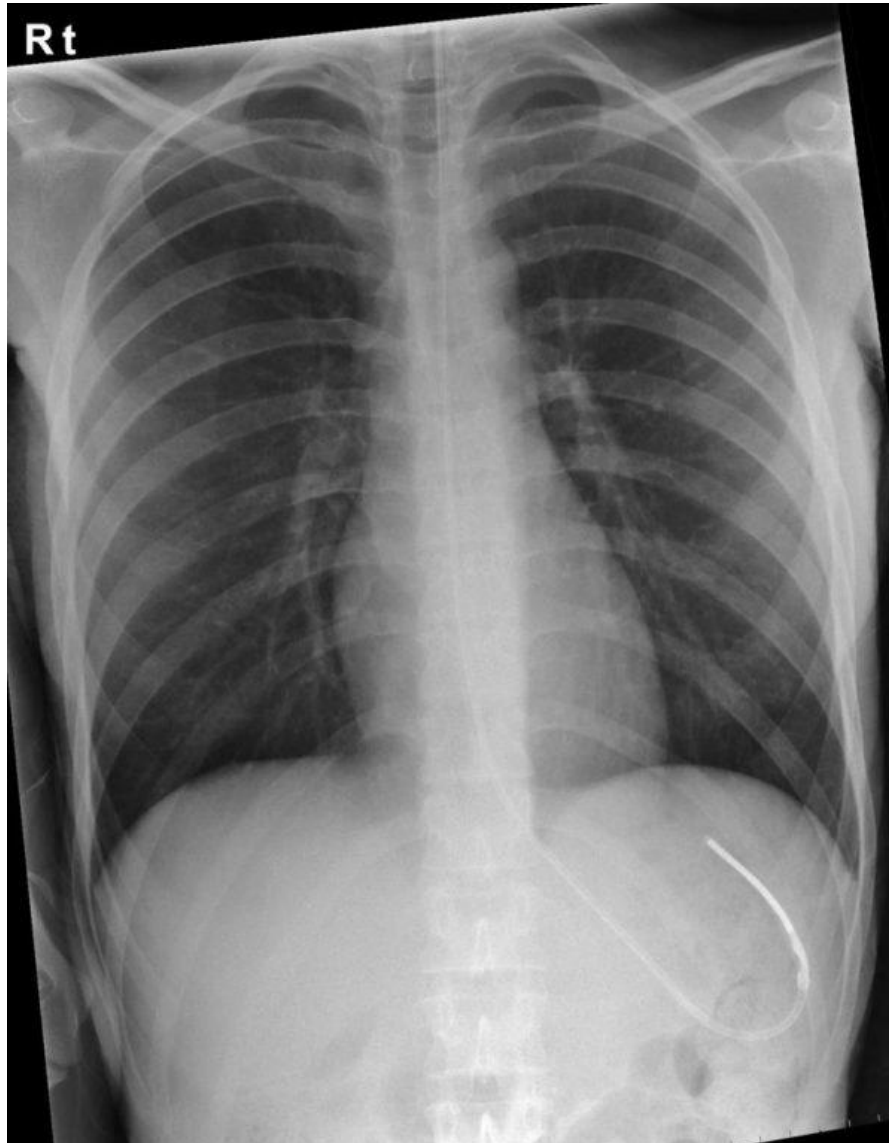
**Left:** air in the gastric fundus as well as free air surrounding the gastric fundus, allowing visualization of both sides of the stomach wall

# Comparison





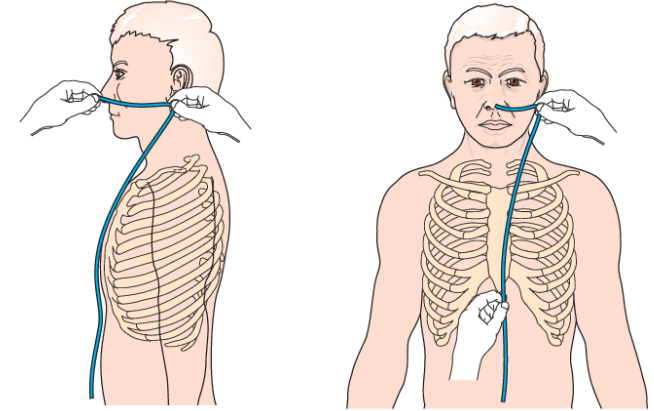
# Nasogastric Tube



[Video here](#)

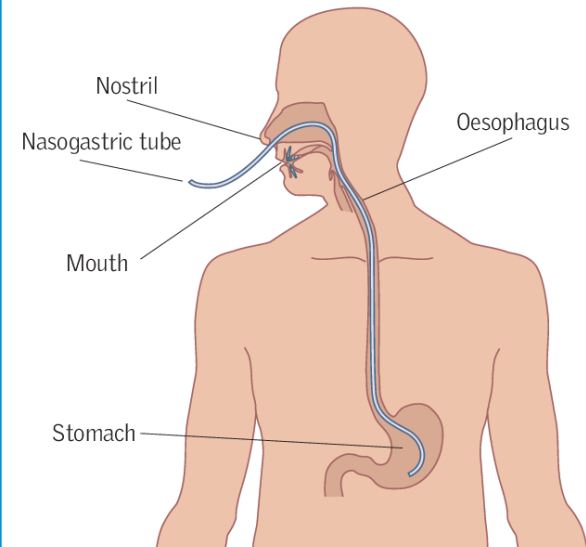
**FIGURE 1**

**Estimating nasogastric tube length**



**FIGURE 1**

**Nasogastric tube position**

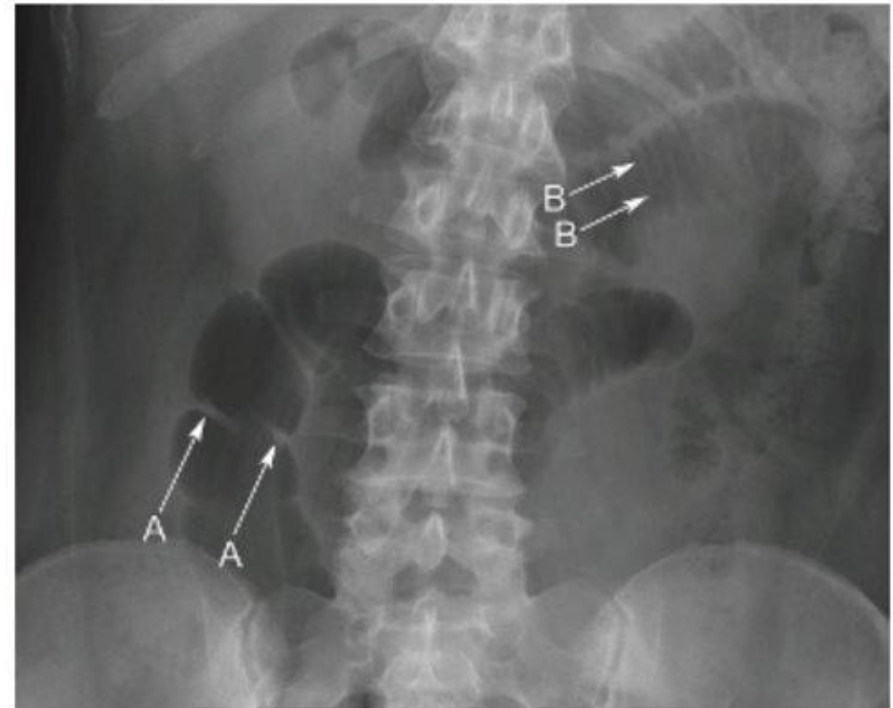
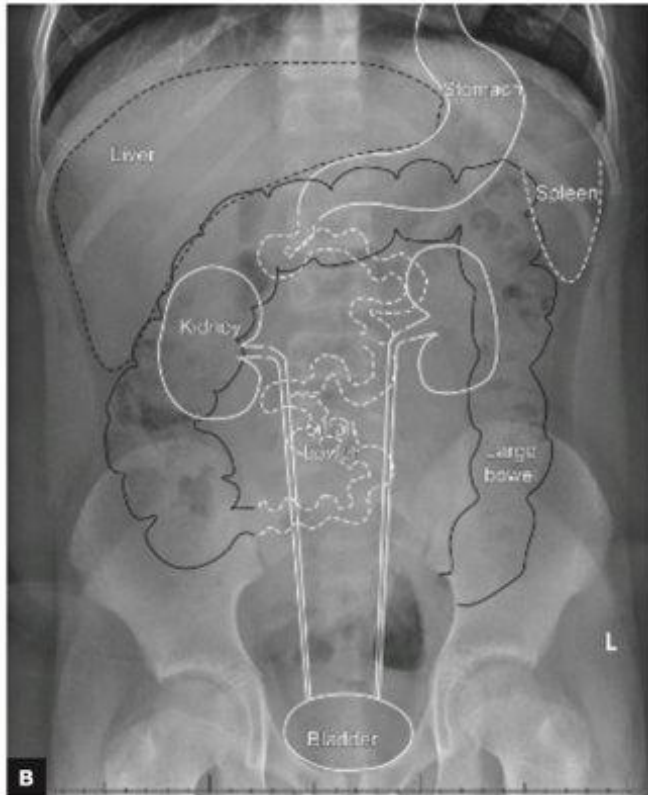


# Structured approach to interpreting Abdominal X-rays

- Important to have a systematic approach to interpreting abdominal X-rays as this decreases the risk of missing pathology.

## BBC approach:

- **Bowel and other organs:**
  - Bowel and bowel gas (too much, too little, distribution, intraluminal, extraluminal, retroperitoneal)
  - Lungs, liver, gallbladder, stomach, psoas muscles (outlines), kidneys, spleen and bladder.
- **Bones:** ribs, lumbar vertebrae, sacrum, coccyx, pelvis and proximal femurs.
- **Calcification and artefact** (e.g., renal stones)



The small bowel's mucosal folds are known as **valvulae conniventes** and are visible across the full width of the bowel (B). The large bowel wall features pouches or sacculations that protrude into the lumen, known as **haustra** (A).

|          | Small bowel          | Large bowel                   |
|----------|----------------------|-------------------------------|
| Position | Central              | Peripheral                    |
| Features | Valvulae conniventes | Haustra                       |
| Size     | <3 cm                | <6 cm or <9 cm in the caecum  |
| Content  | Liquid and air       | Solid faeces, liquid, and air |

## Case II

A 46-year-old woman comes to the ED because of abdominal pain and vomiting. She is unable to keep any fluids down and is only passing small amounts of flatus. Her past medical history includes two Caesarean sections and a laparoscopic sterilization. Examination shows that she is afebrile, with a HR 100 bpm and a BP of 120/80 mmHg. Physical exam shows abdominal distension, general tenderness, and loud, tinkling bowel sounds. Abdominal surgical scars are visible, consistent with previous Caesarean section and laparoscopic surgery history.



Fig. 4.96

This radiograph of the abdomen, anteroposterior view, demonstrates a number of dilated loops of small bowel. Small bowel can be identified by the plicae circulares that pass from wall to wall as indicated. The large bowel is not dilated. The cause of the small bowel dilatation is an adhesion after pelvic surgery.

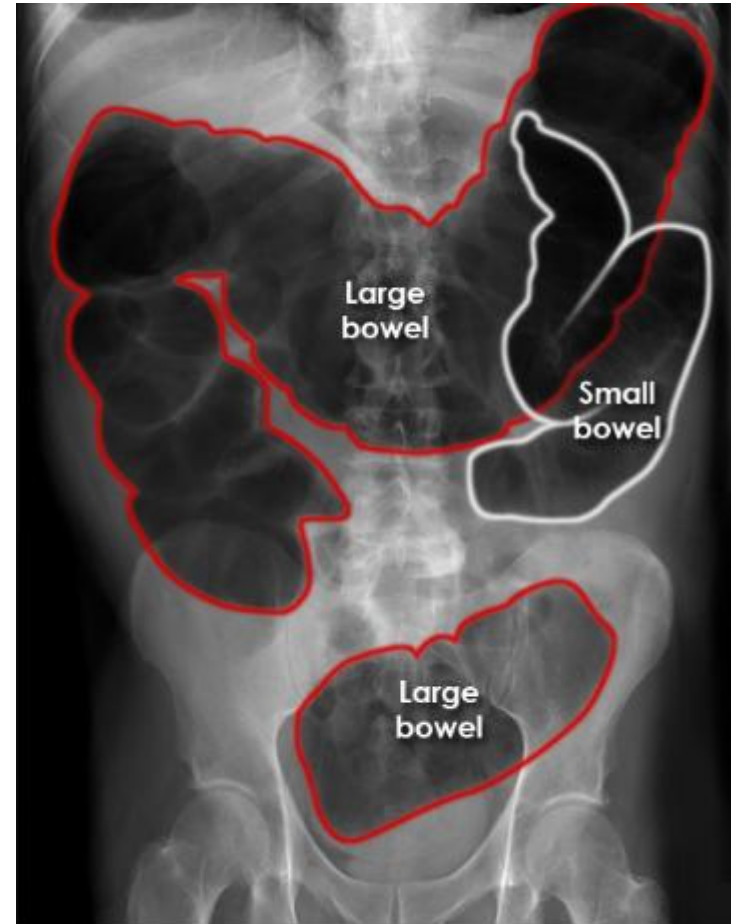
# Small vs. Large Bowel Obstruction



*(previously covered – see prior lecture slides)*



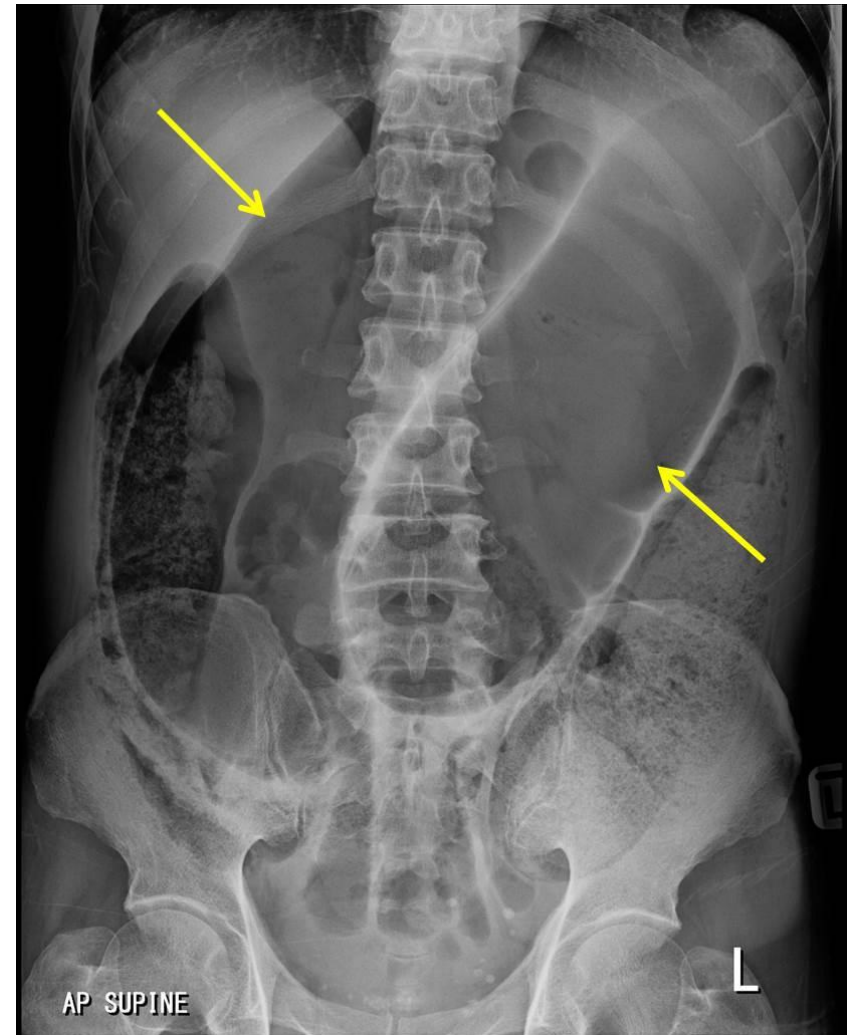
# Can occur simultaneously



[Abdominal X-ray - Abnormalities - Large bowel obstruction](#)

## Case III

A 72-year-old woman comes to the emergency department because of a 10-day history of worsening, generalized, colicky abdominal pain and vomiting. She has not defecated for 4 days and is now unable to pass flatus (obstipated). She is afebrile, mildly tachycardic (HR 110 bpm), and mildly tachypneic (respiratory rate 24 bpm). Her BP is 160/90 mmHg. Clinical examination shows a grossly distended abdomen, generalized abdominal tenderness on palpation, and tinkling bowel sounds. Blood tests show mild leukocytosis and mildly elevated CRP. Supine AXR is performed.

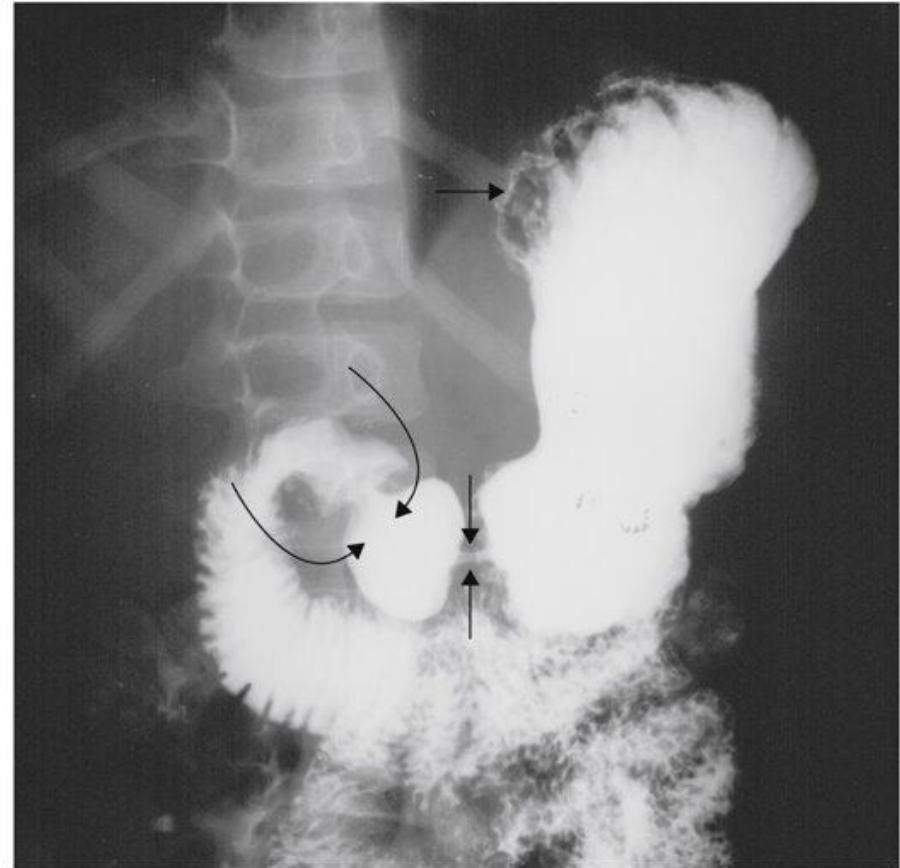


**Sigmoid Volvulus:** sigmoid colon extends out of left iliac fossa into upper quadrants

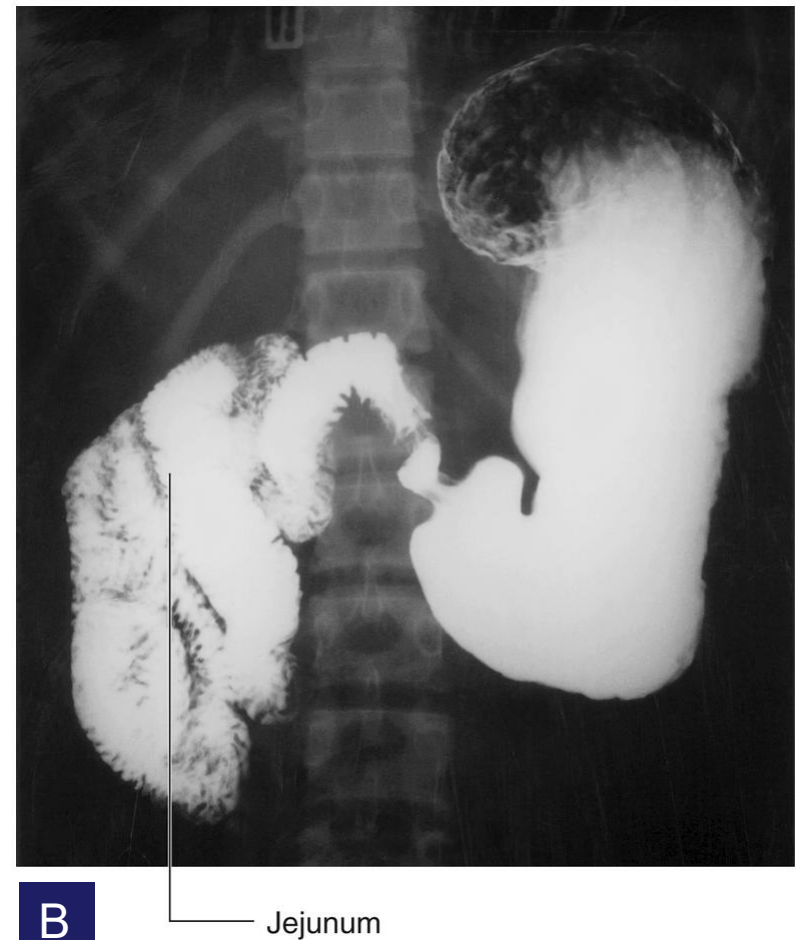
# Gastrointestinal Contrast Studies

- For evaluation of the upper GI tract (esophagus, stomach, and duodenum); endoscopy is more commonly performed.  
**Upper GI contrast studies are still accurate and safe.**

- Normal upper GI series.**  
**Barium-filled stomach and duodenum.** The patient is in the prone position.
  - Gas** (*horizontal arrow*) is seen in the gastric fundus;
  - Peristaltic wave** (*vertical arrows*) crosses the gastric antrum;
  - Pylorus** (*curved arrows*) separates the duodenal bulb and stomach.







**A:** Normal radiograph of jejunum and ileum

**B:** Small Bowel Malrotation. Radiograph of stomach, duodenum, and jejunum using barium

# Large Bowel Contrast Study

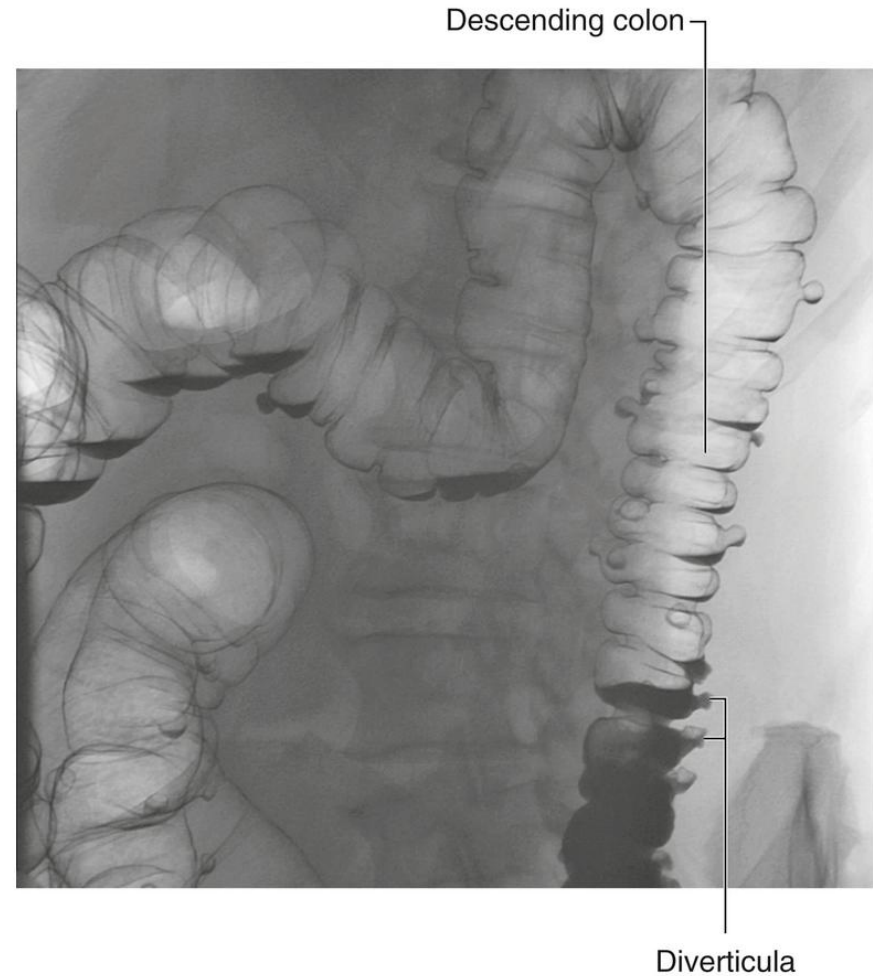
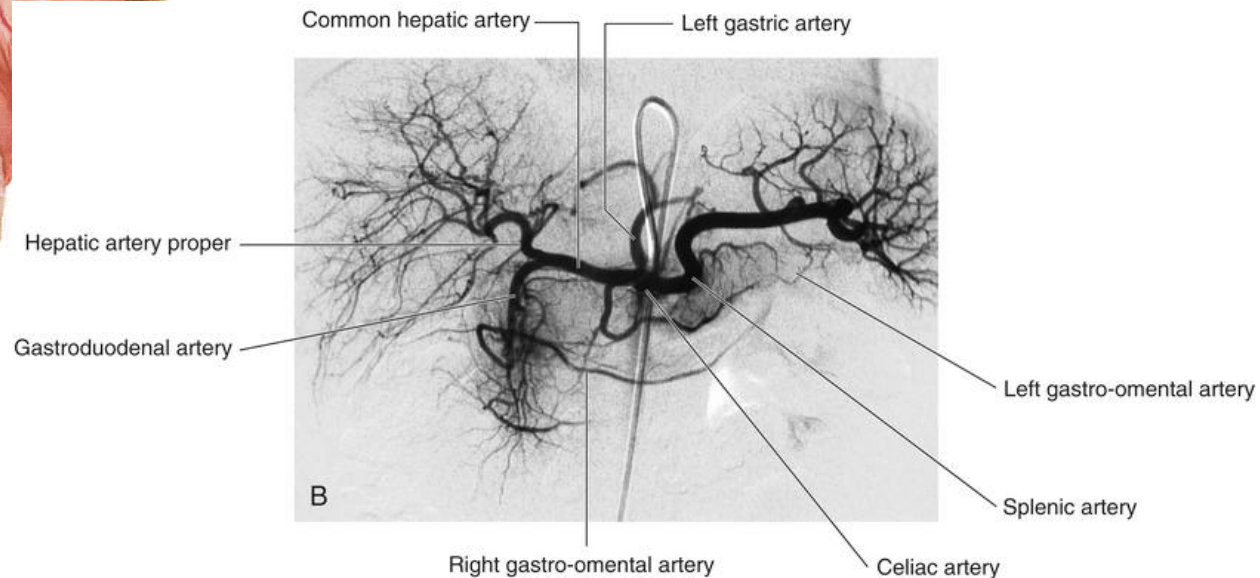
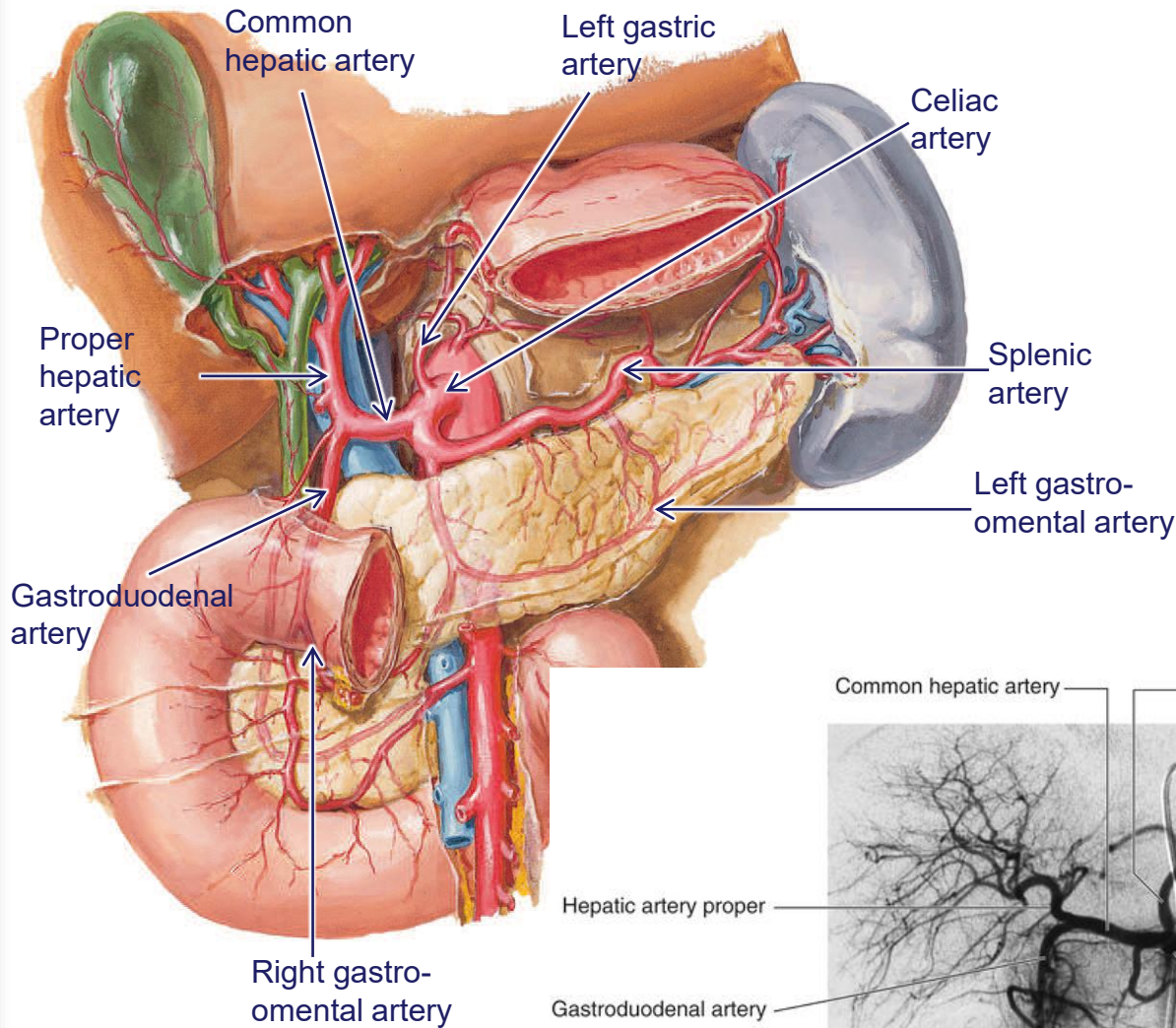


Fig. 4.100

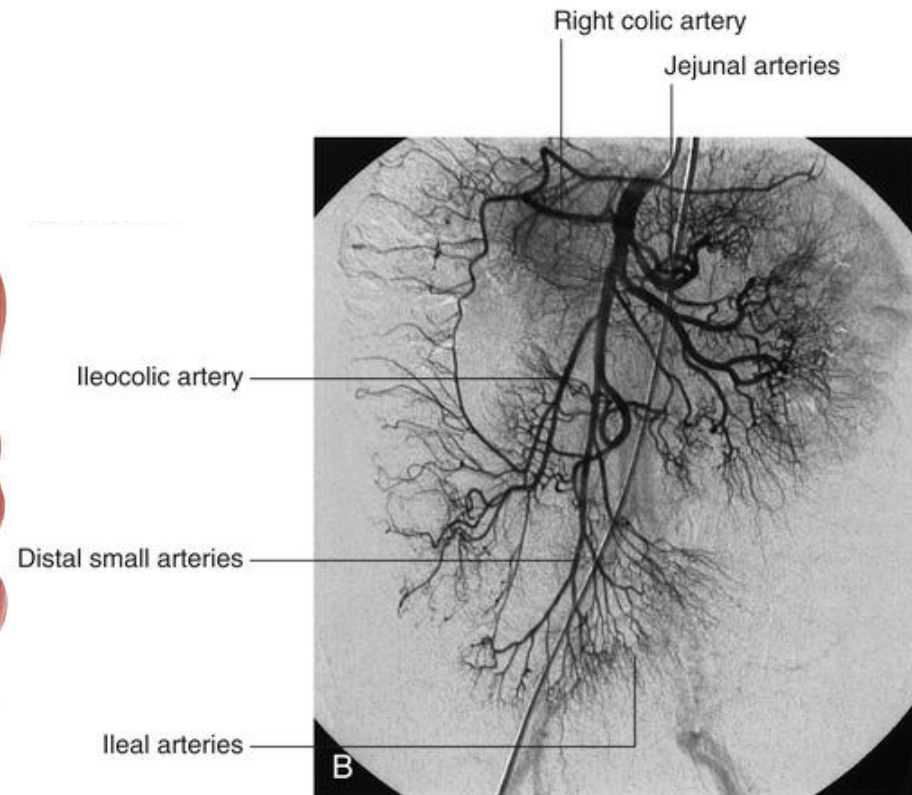
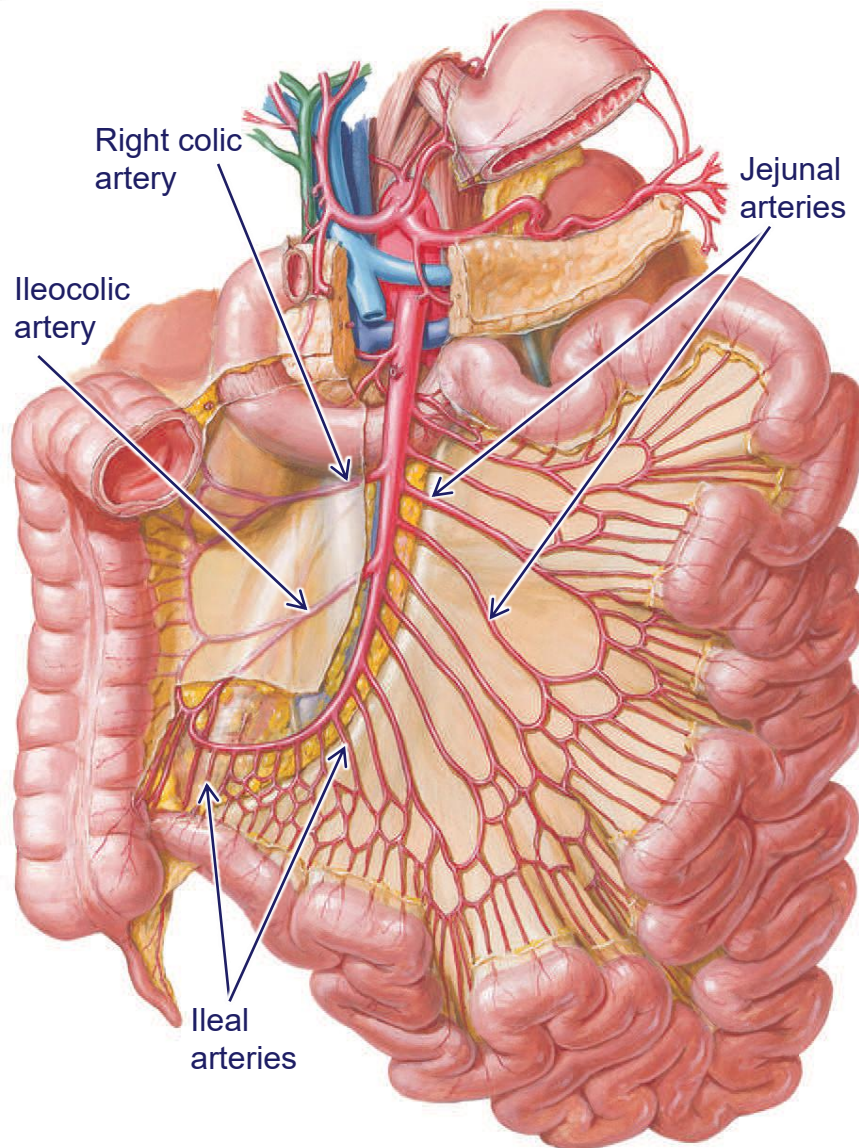
This double-contrast barium enema demonstrates numerous small outpouchings throughout the distal large bowel predominantly within the descending colon and the sigmoid colon. These small outpouchings are diverticula and, in most instances, remain quiescent.

# Contrast Study - Angiography

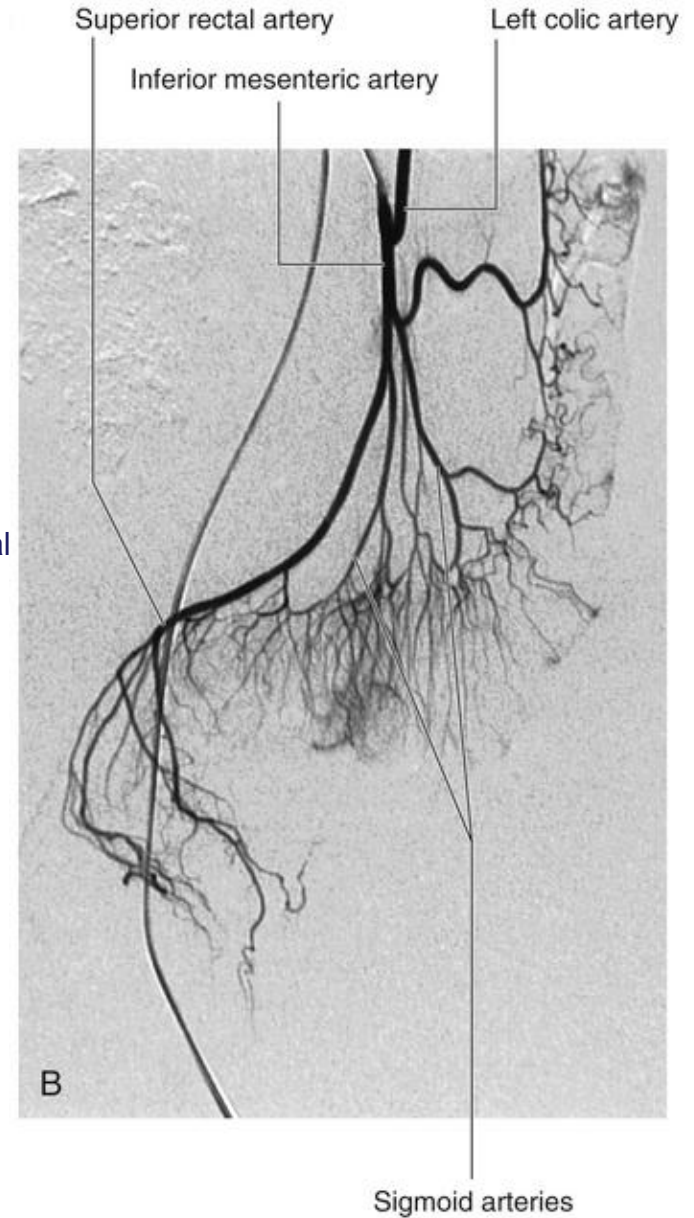
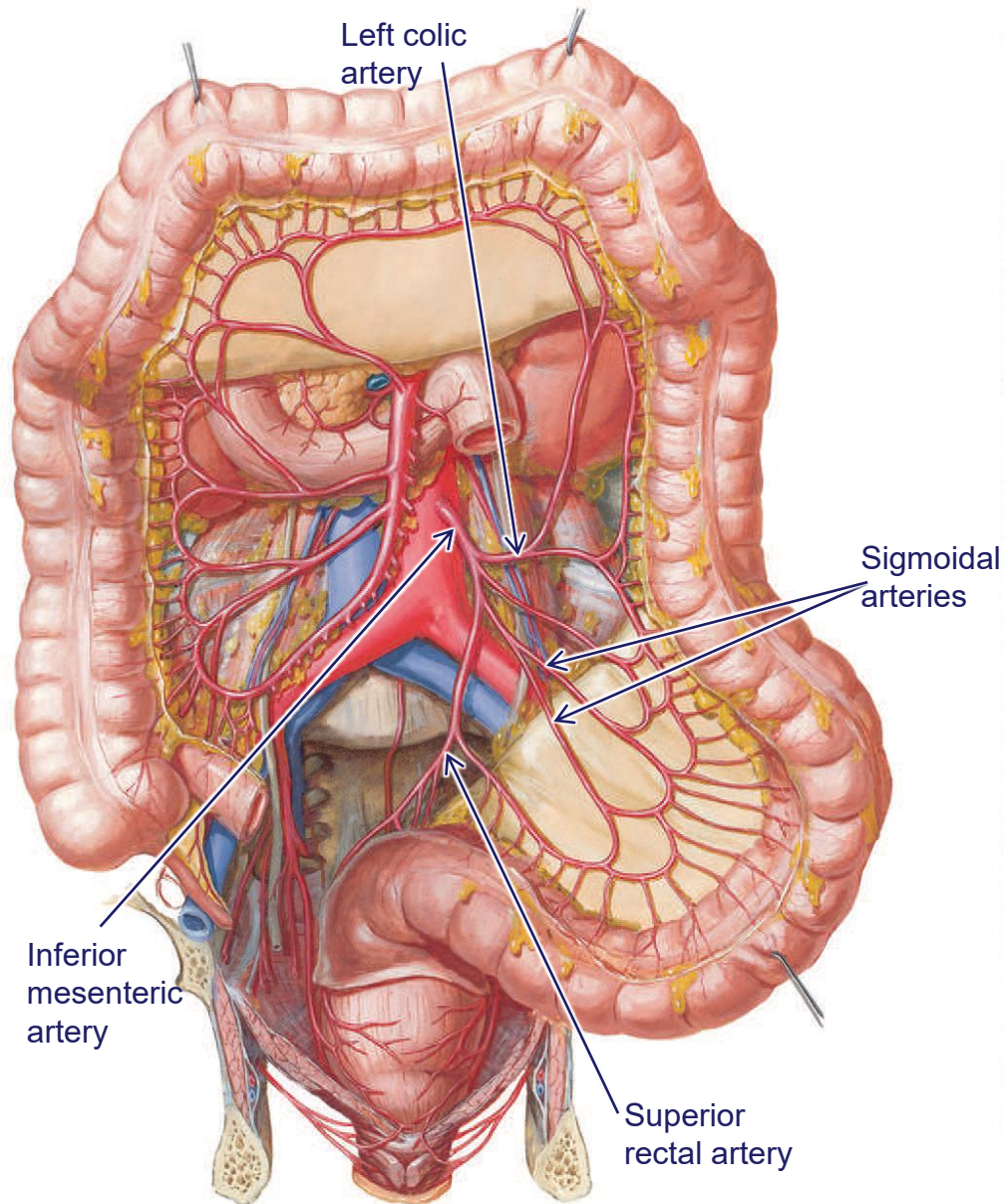




# Contrast Study - Angiography



# Contrast Study - Angiography





# Computed Tomography

Help identify the cause of abdominal or pelvic pain, diagnose diseases of the internal organs, small bowel and colon, such as:

- Infections like appendicitis, pyelonephritis or infected fluid collections, also known as abscesses.
- Inflammatory bowel disease such as ulcerative colitis or Crohn's disease, pancreatitis or liver cirrhosis.
- Cancers of the liver, kidneys, pancreas, ovaries and bladder as well as lymphoma.
- Kidney and bladder stones.
- Abdominal aortic aneurysms (AAA), injuries to abdominal organs such as the spleen, liver, kidneys, or other internal organs in cases of trauma.
- Guide biopsies and other procedures such as abscess drainages and minimally invasive tumor treatments.
- Plan for and assess the results of surgery, such as organ transplants.
- Stage, plan and properly administer radiation treatments for tumors as well as monitor response to chemotherapy.

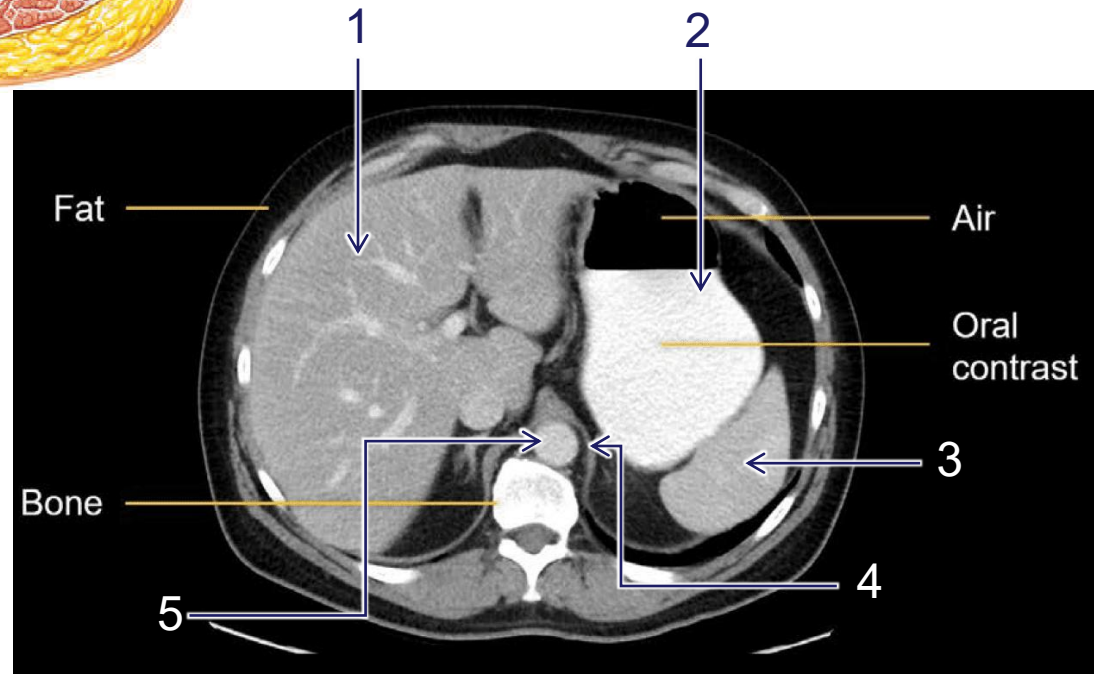
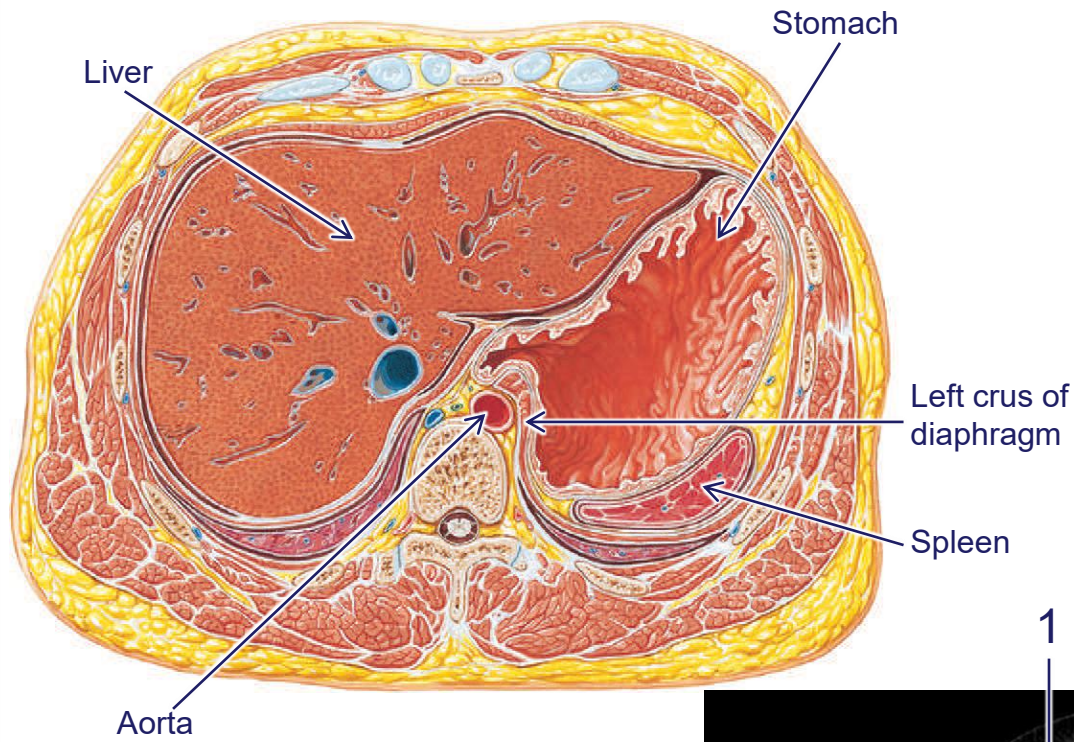


# Computed Tomography

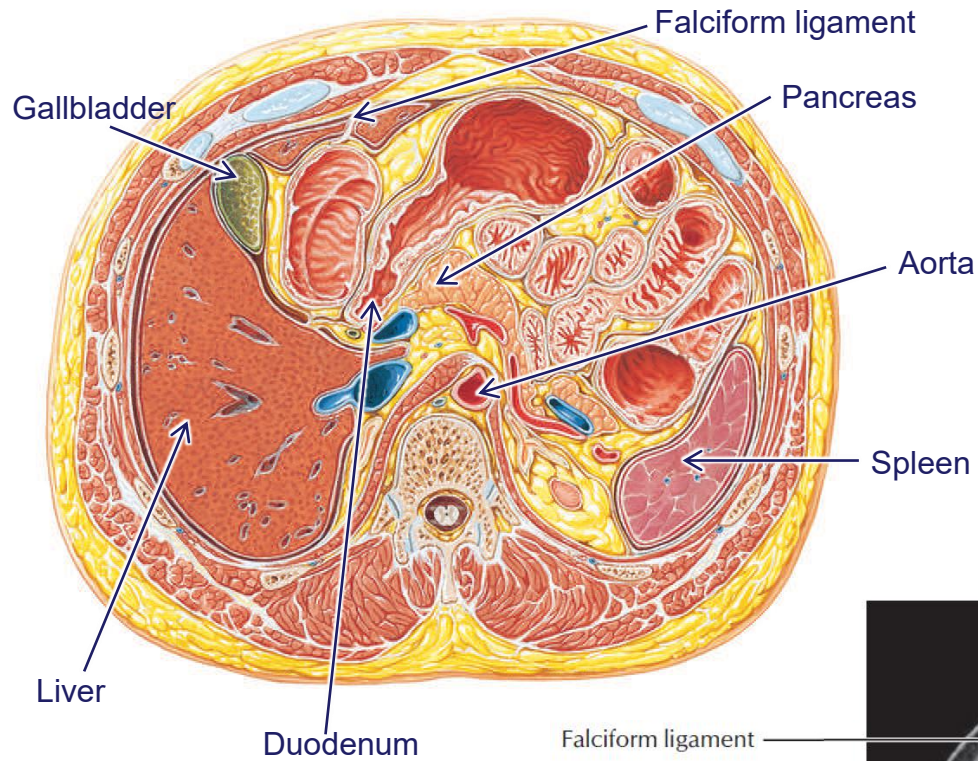
- Images are still in shades of gray
- **BONE and Metal IS BRIGHT WHITE!**
- Better definition of soft tissue
- Enhances the appearance of traumatic injuries that were not visible on X-Ray
- Images can be enhanced with window views
- Manipulation of grey scale to enhance structures
- Now image reconstruction can create 3D Images
- Can utilize **contrasts**
- CT Scan without Contrast vs. CT Scan with Contrast

|                                                                                                          |                                                                                                         |
|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|  <p>COMPACT BONE</p>  |  <p>Bright white</p> |
|  <p>MUSCLE</p>        |                                                                                                         |
|  <p>FLUID</p>         |                                                                                                         |
|  <p>FAT</p>          |                                                                                                         |
|  <p>BONE MARROW</p> |                                                                                                         |

# Axial CT Scan



# Axial CT Scan



Falciform ligament

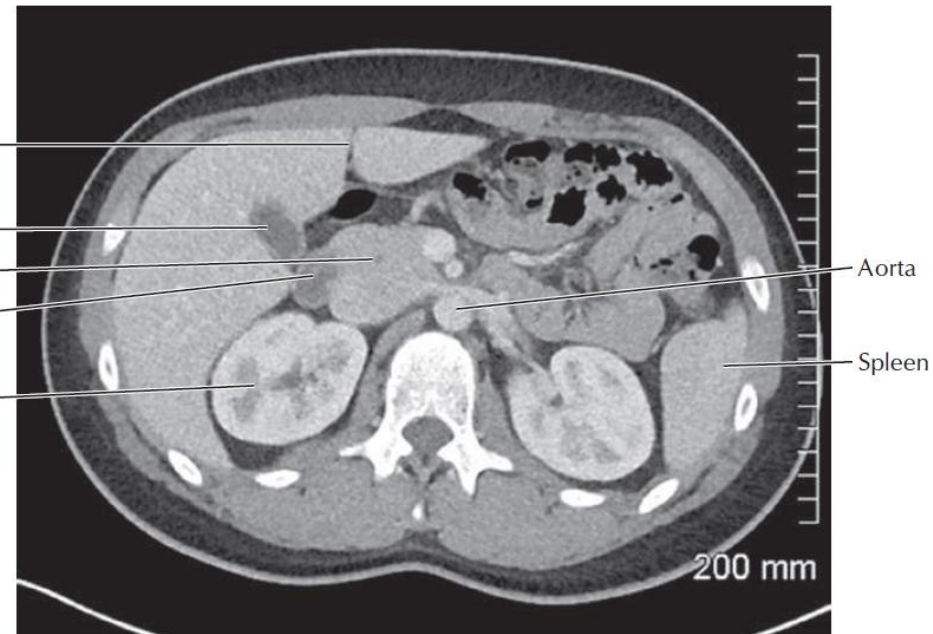
Gallbladder

Head of pancreas

Duodenum

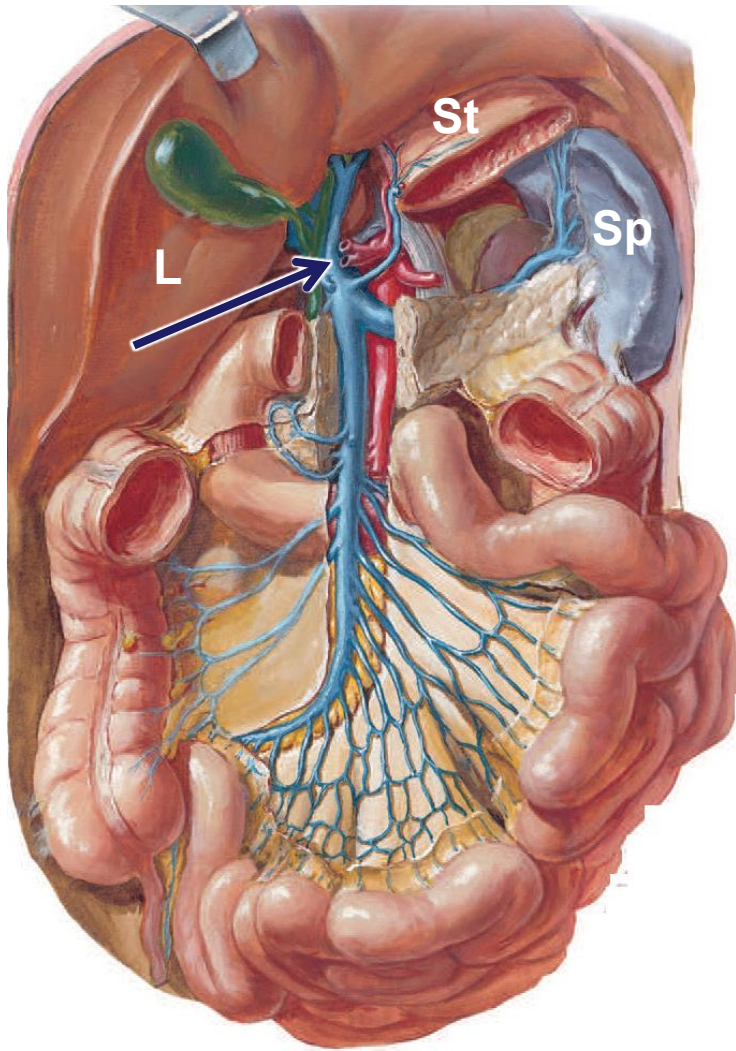
Right kidney

**Axial CT image of  
midabdomen with  
intravenous contrast  
enhancement**





# Coronal CT



L – Liver, St – Stomach, SP – Spleen, Arrow – Portal vein

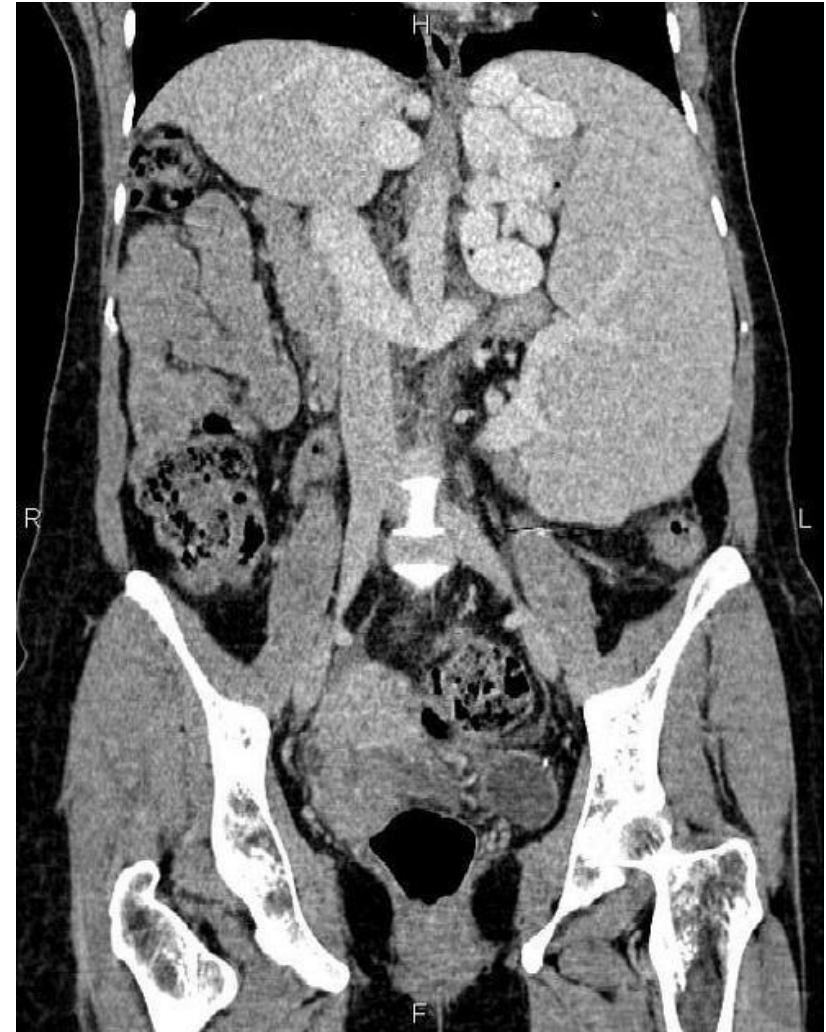


# Compare – Coronal CT



Normal

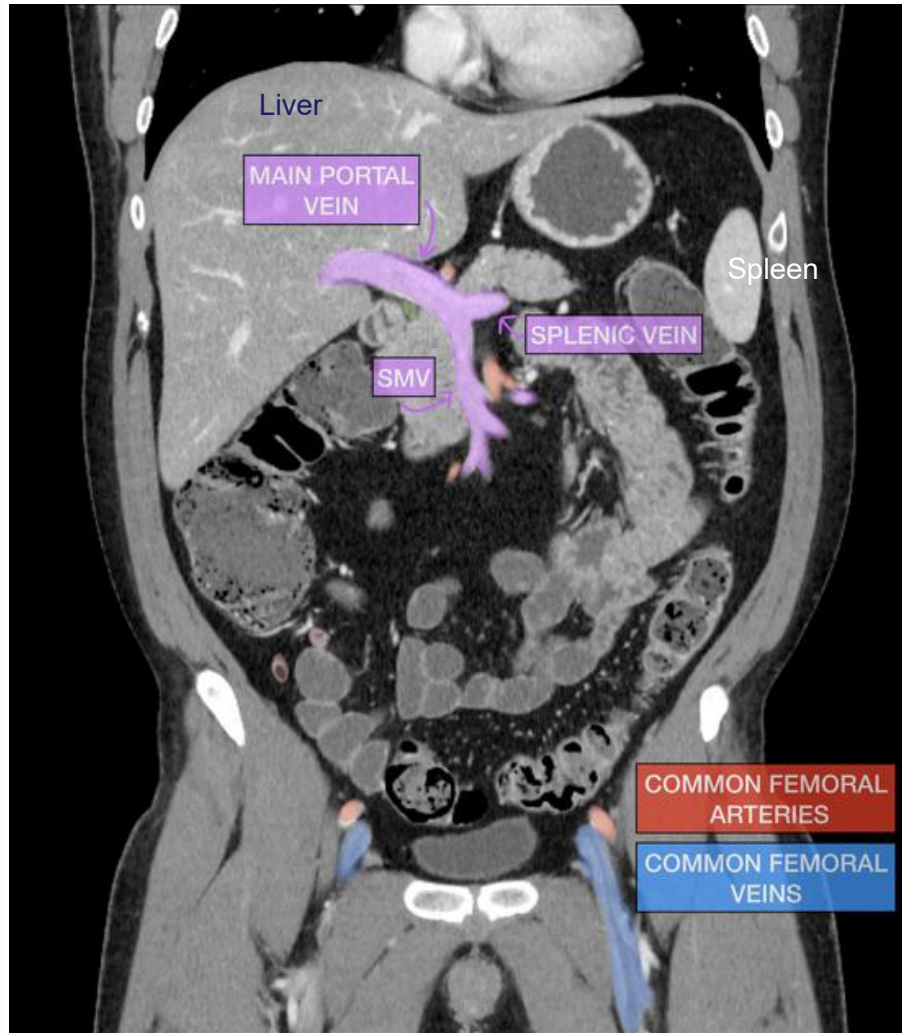
Which vein(s) are colored?



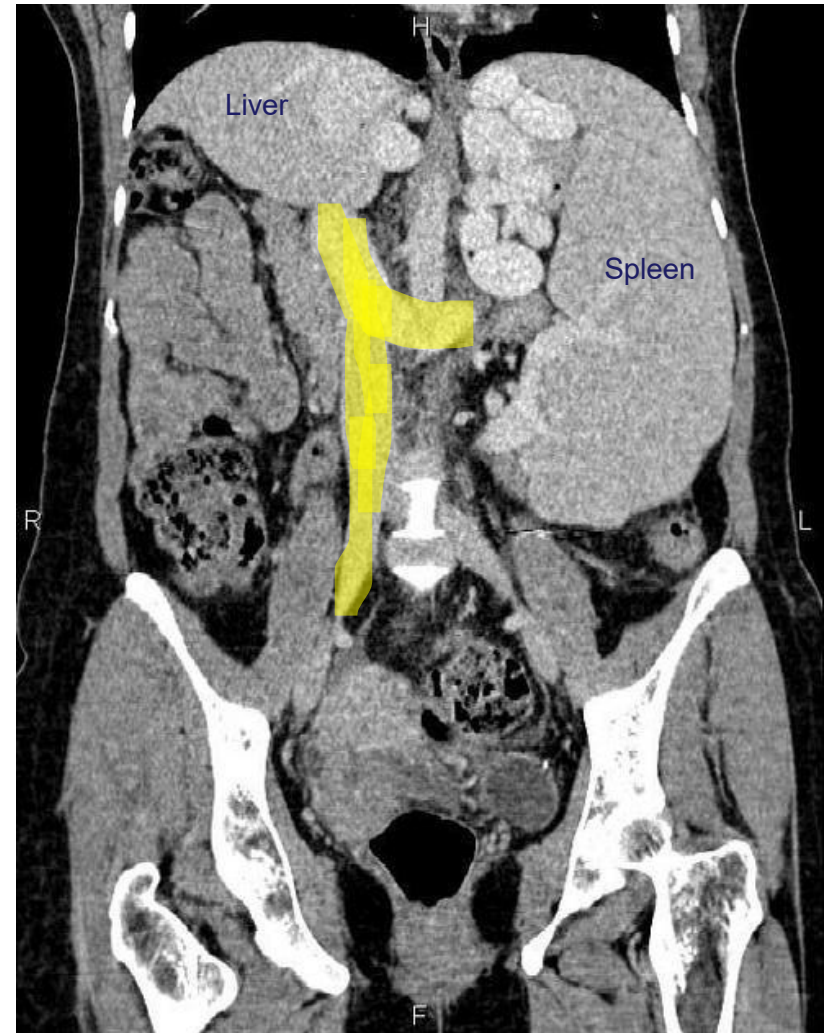
Patient

Assess the liver and spleen and veins

# Compare – Coronal CT



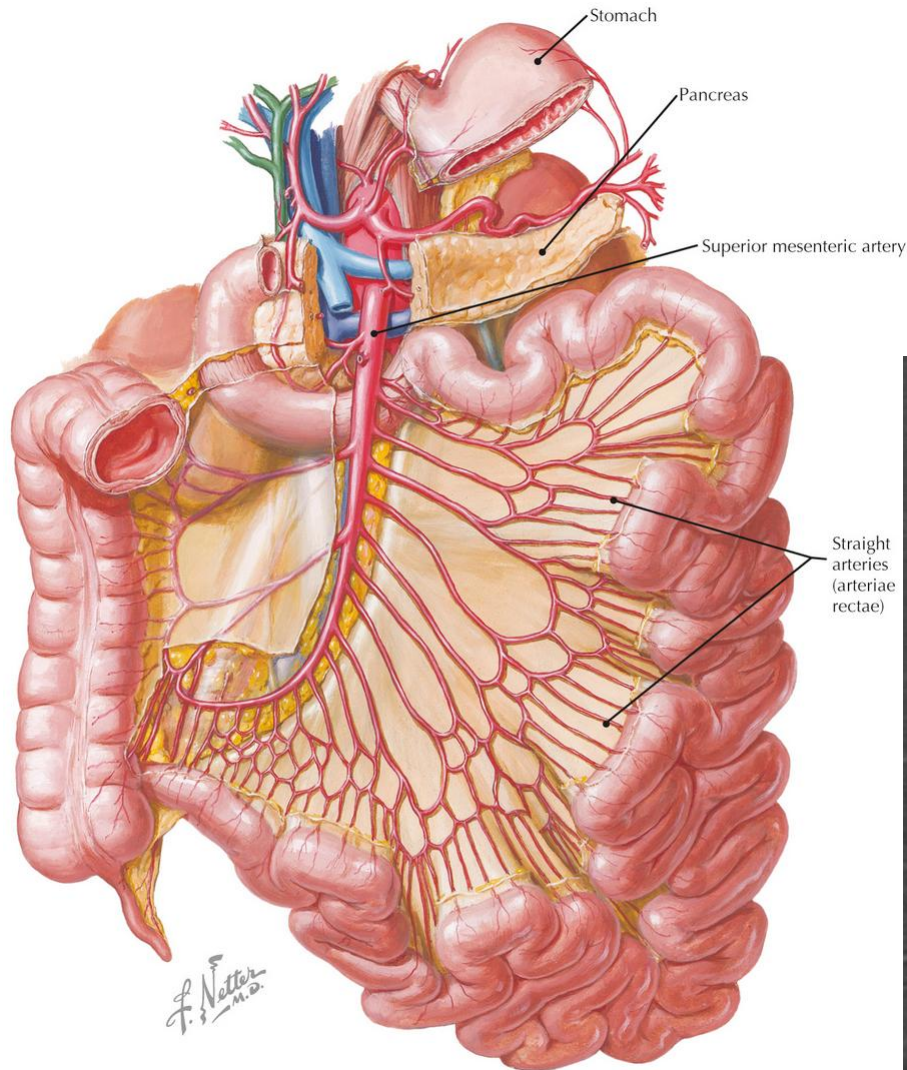
Normal



Patient  
Assess the liver and spleen  
and veins



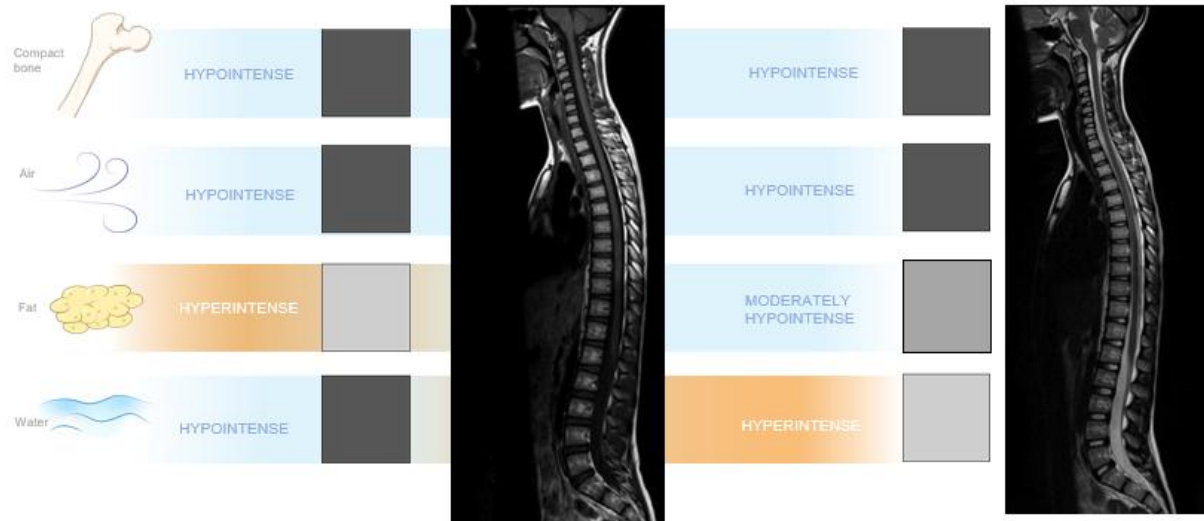
# Contrast CT - Arteries of the Small Bowel



High-Value Multidetector CT Angiography of the Superior Mesenteric Artery: What Emergency Medicine Physicians and Interventional Radiologists Need to Know | RadioGraphics (rsna.org)

# Magnetic Resonance Imaging (MRI)

- Complementary to other imaging modalities (CT, US)
- Liver, biliary tree, spleen, pancreas, kidneys, vessels
- Allows for assessment of inflammatory bowel disease and cancer
- Particularly useful in young patients with acute/chronic disorders requiring short-term follow up (no radiation)
- Limitations:
  - Normal bowel motility (motion artifacts)
  - High cost, limited accessibility

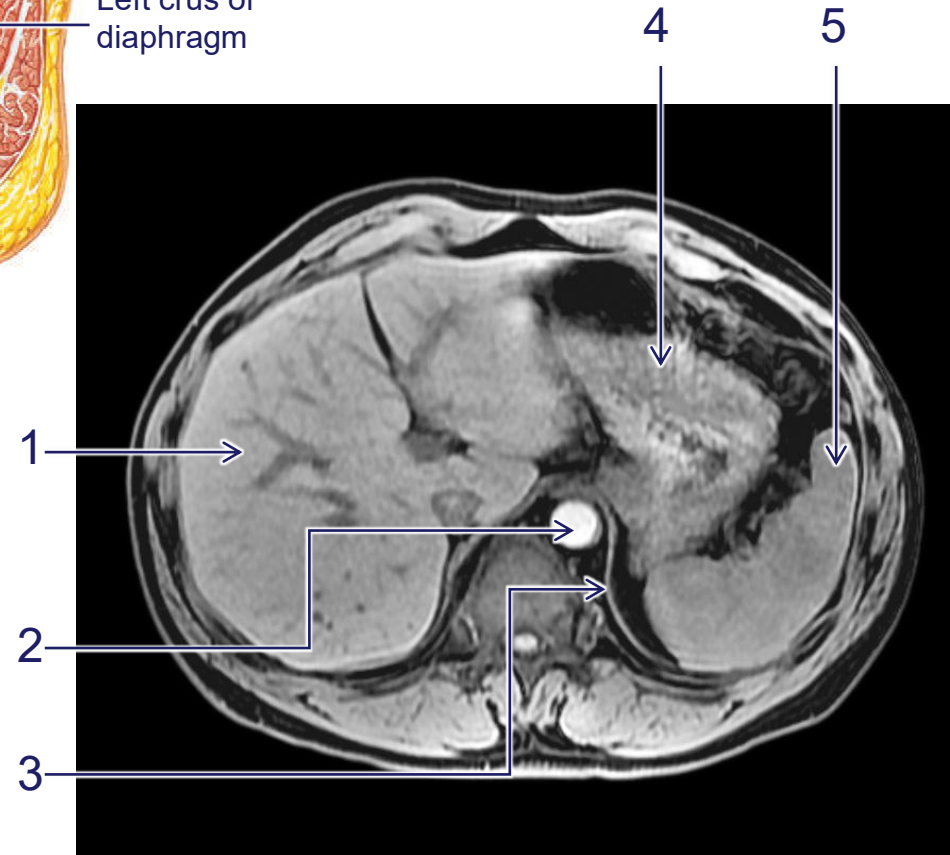
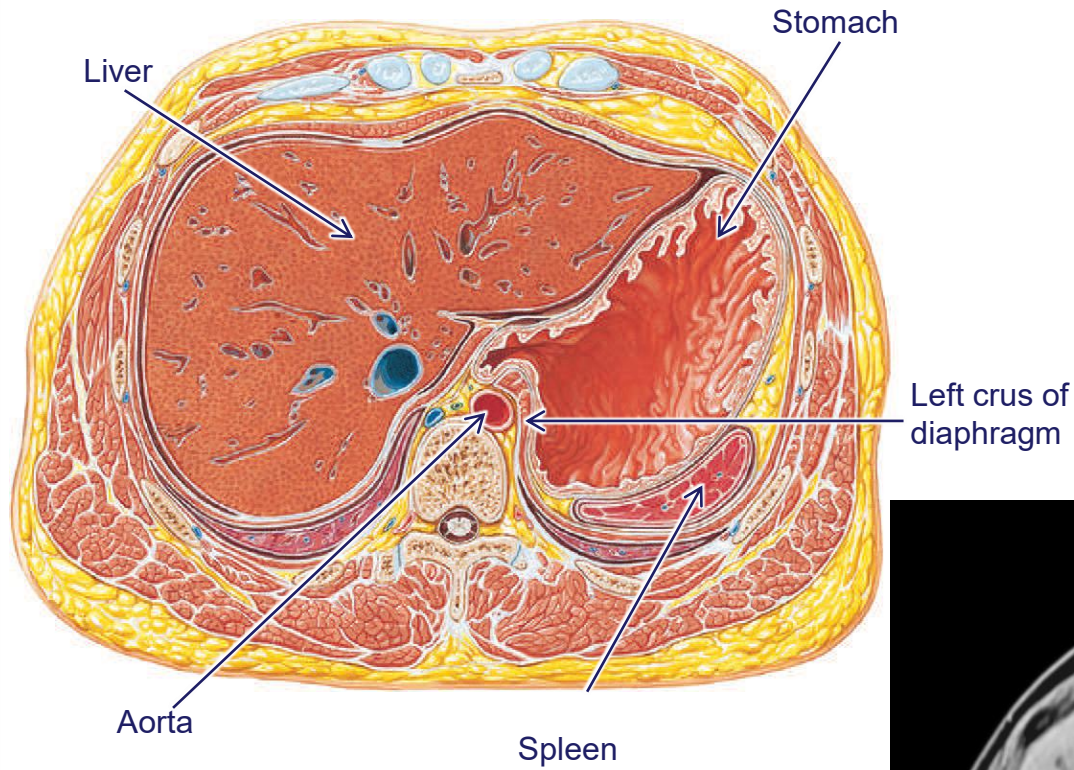


Sagittal MRI of the spine. Note the dark cerebrospinal fluid (CSF)

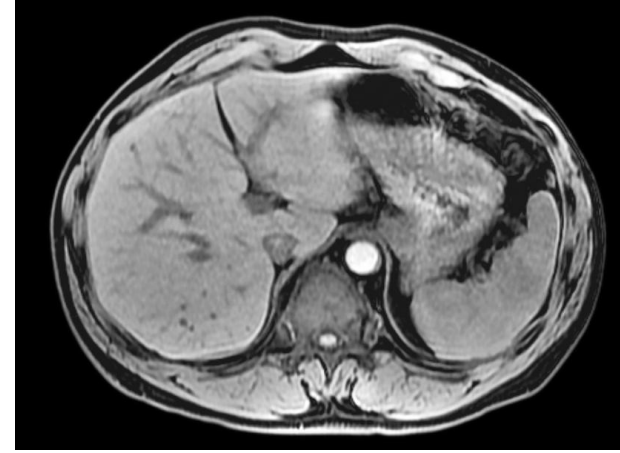
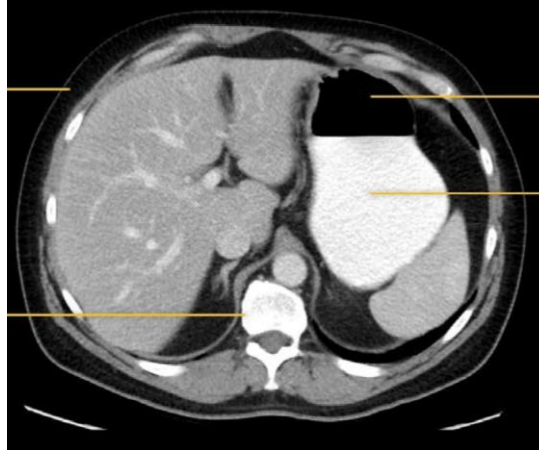
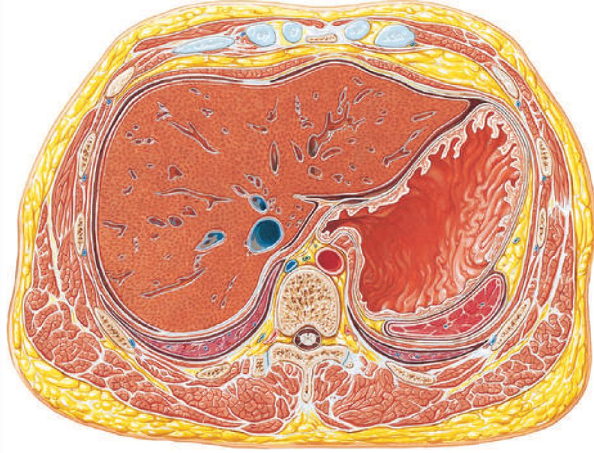
Sagittal MRI of the spine. Note the bright cerebrospinal fluid (CSF)



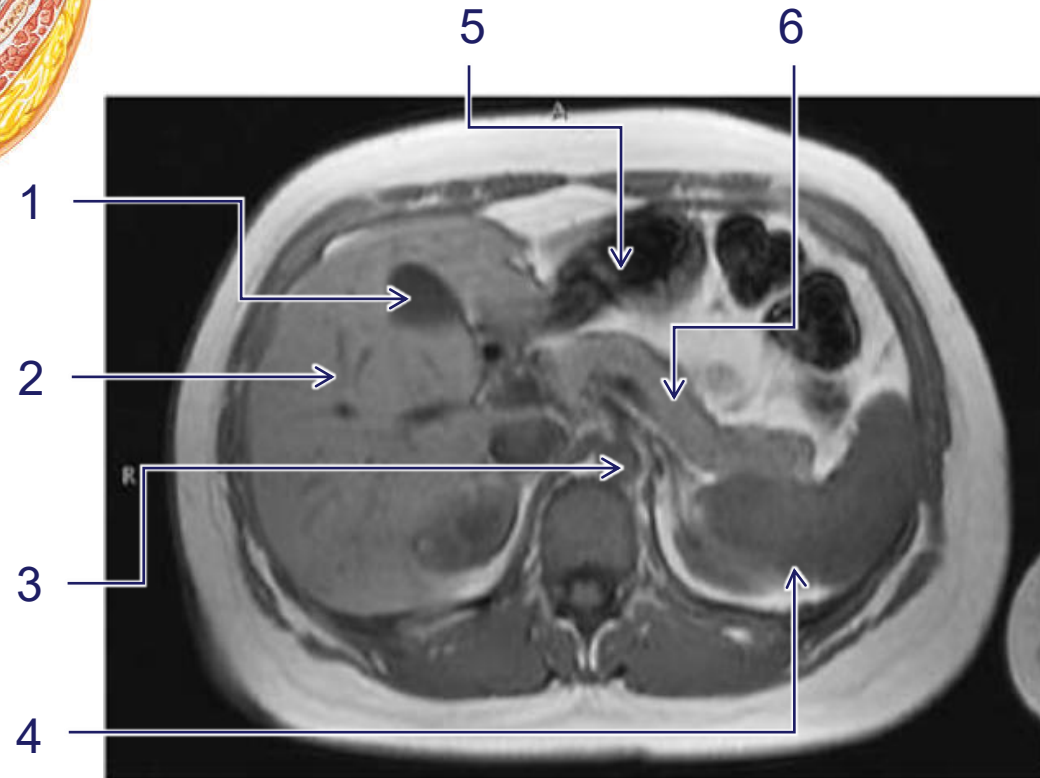
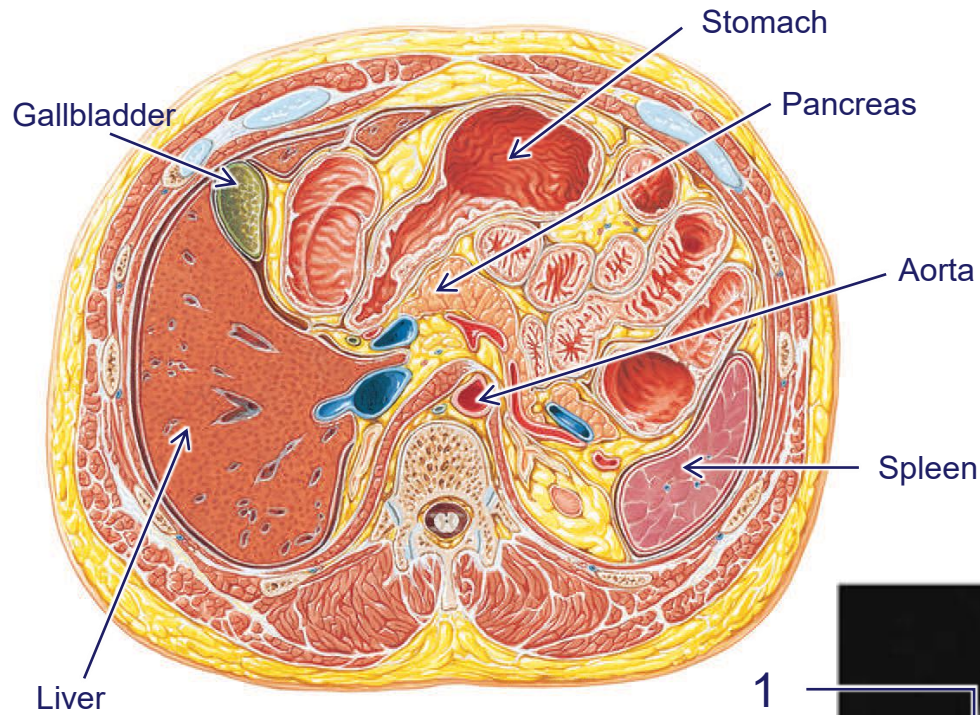
# Axial MRI



# Comparison: CT - MRI



# Axial MRI





# Magnetic Resonance Cholangiopancreatography (MRCP)

